Egg Drops: Is a 1-3 or a 2-2 Strategy Better?

A constant-factor comparison for four eggs and N floors

Question. With four eggs and N floors, compare:

- **1-3:** use one egg for a coarse jump search; after it breaks, solve the remaining interval with the other three eggs.
- **2-2:** divide the building into blocks of F floors; solve the coarse problem on the N/F blocks with two eggs, and then solve the critical block with the other two eggs.

Both have order $N^{1/4}$ drops. The issue is the constant.

1. The two-egg and three-egg subproblems

The 2026 talk emphasizes that worst-case performance is the quantity to minimize, and it first develops the one-egg and two-egg cases. It also points out the key improvement for two eggs: instead of using a constant step size, use decreasing steps $x, x-1, x-2, \ldots$. For example, $105 = 14+13+\cdots+1$, so 14 drops suffice. This is the triangular-number strategy.[1]

There is also an exact counting fact in the accompanying paper: with k eggs and d drops, the number of floors that can be handled is

$$\sum_{j=1}^{k} \binom{d}{j}.$$

This is the standard dynamic egg-drop count and gives the true asymptotic constants for the two- and three-egg subproblems.

1.1. Two eggs

For two eggs, the capacity after d drops is

$$\binom{d}{1} + \binom{d}{2} = d + \frac{d(d-1)}{2} = \frac{d(d+1)}{2}.$$

Therefore the minimum number of drops for m floors is

$$T_2(m) = \left\lceil \frac{\sqrt{8m+1}-1}{2} \right\rceil.$$

In particular,

$$T_2(m) = \sqrt{2m} + O(1).$$

Thus the leading constant for the two-egg problem is $\sqrt{2}$.

[1] The talk's slides explicitly describe the decreasing-step method and the 105-floor example. See the cited source links at the end.

1.2. Three eggs

For three eggs, the capacity after d drops is

$$\binom{d}{1} + \binom{d}{2} + \binom{d}{3} = \frac{d^3 + 5d}{6}.$$

Hence the minimum number of drops $T_3(m)$ satisfies

$$\frac{d^3 + 5d}{6} \geq m,$$

and therefore

$$T_3(m) = (6m)^{1/3} + O(1).$$

So the leading constant for the three-egg problem is $6^{1/3}$.

Important distinction. The paper's main one-dimensional theorem proves the simpler fixed-step recursive bound

$$P_1(k) \leq \left\lceil kN^{1/k} \right\rceil.$$

For $k = 2$ this gives the constant 2, rather than the sharper $\sqrt{2}$ obtained from the triangular/dynamic two-egg strategy. The paper explicitly says its fixed-step strategy is not optimal, while noting that the triangular strategy is slightly better in the two-egg case.[2]

2. The 1-3 strategy

Let S be the size of the coarse jump. We drop the first egg at floors

$$S,\ 2S,\ 3S, \ldots$$

until it breaks. In the worst case this costs approximately

$$\frac{N}{S}$$

drops. Once the first egg breaks, the unknown critical floor lies in an interval of at most S floors, which is now a three-egg problem.

Using the exact asymptotic for the three-egg problem, the total worst-case cost is

$$f_{13}(S) = \frac{N}{S} + (6S)^{1/3} + O(1).$$

The $O(1)$ term records the harmless rounding/discreteness effects; they do not affect the leading constant.

[2] See the discussion surrounding the two-egg strategy and the paper's remark that its uniform-step method is not optimal.

2.1. Optimize the coarse step

Ignoring the lower-order $O(1)$ term, differentiate:

$$f'_{13}(S) = -\frac{N}{S^2} + \frac{6^{1/3}}{3} S^{-2/3}.$$

Set this equal to zero:

$$\frac{N}{S^2} = \frac{6^{1/3}}{3} S^{-2/3}.$$

Multiplying by S^2 gives

$$N = \frac{6^{1/3}}{3} S^{4/3},$$

so

$$S_* = \left(\frac{3N}{6^{1/3}}\right)^{3/4} = \frac{3^{3/4}}{6^{1/4}} N^{3/4}.$$

At the optimum, it is convenient to use the balance equation. From

$$N = \frac{6^{1/3}}{3} S_*^{4/3},$$

we get

$$\frac{N}{S_*} = \frac{6^{1/3}}{3} S_*^{1/3}.$$

Thus the two terms are in a fixed ratio of $1 : 3$, and

$$f_{13}(S_*) = \frac{4}{3}\, 6^{1/3} S_*^{1/3}.$$

Substituting S_* gives

$$\boxed{f_{13}(S_*) \sim 4 \left(\frac{2}{9}\right)^{1/4} N^{1/4}}$$

with numerical constant

$$4 \left(\frac{2}{9}\right)^{1/4} \approx 2.74636.$$

So the 1-3 construction, when its three-egg stage is itself solved optimally, costs

$$\boxed{2.74636\, N^{1/4} + O(1)}.$$

3. The 2-2 strategy

Now partition the N floors into blocks of F floors. There are about N/F blocks.

Use two eggs to identify the critical block. This costs asymptotically

$$T_2(N/F) \sim \sqrt{\frac{2N}{F}}.$$

Then use the other two eggs to locate the critical floor inside the chosen block. This costs

$$T_2(F) \sim \sqrt{2F}.$$

Therefore

$$f_{22}(F) = \sqrt{\frac{2N}{F}} + \sqrt{2F} + O(1).$$

Set $y = \sqrt{F}$. Then

$$f_{22}(y) = \sqrt{2}\left(\frac{\sqrt{N}}{y} + y\right) + O(1).$$

Differentiate:

$$f'_{22}(y) = \sqrt{2}\left(-\frac{\sqrt{N}}{y^2} + 1\right).$$

The critical point is

$$y^2 = \sqrt{N}, \qquad \text{so} \qquad \boxed{F_* = \sqrt{N}}.$$

At this choice,

$$f_{22}(F_*) = \sqrt{2}\left(N^{1/4} + N^{1/4}\right) + O(1),$$

so

$$\boxed{f_{22}(F_*) \sim 2\sqrt{2}\, N^{1/4}}$$

with numerical constant

$$2\sqrt{2} \approx 2.82843.$$

4. Which is better?

We have found

$$C_{13} = 4\left(\frac{2}{9}\right)^{1/4} \approx 2.74636, \qquad C_{22} = 2\sqrt{2} \approx 2.82843.$$

Therefore

$$C_{13} < C_{22}.$$

The ratio is

$$\frac{C_{13}}{C_{22}} \approx 0.971.$$

So asymptotically the 1-3 strategy uses about 2.9% fewer drops than the 2-2 strategy.

Strategy	Optimal scale	Leading constant in $N^{1/4}$
1-3	$S \asymp N^{3/4}$	$4(2/9)^{1/4} \approx 2.74636$
2-2	$F \asymp N^{1/2}$	$2\sqrt{2} \approx 2.82843$

5. A useful warning about what is being compared

There are two different levels of approximation in the talk and paper, and keeping them separate resolves an apparent contradiction.

If both constructions use the paper's uniform fixed-step bound

The paper proves

$$P_1(k) \le kN^{1/k} + O(1).$$

Using that bound for the three-egg subproblem gives

$$f_{13}(S) \approx \frac{N}{S} + 3S^{1/3},$$

whose minimum is $4N^{1/4}$. If the two-egg subproblem in the 2-2 construction is also estimated by the paper's fixed-step bound $2\sqrt{m}$, then

$$f_{22}(F) \approx 2\sqrt{N/F} + 2\sqrt{F},$$

whose minimum is also $4N^{1/4}$.

Thus, *at the level of the paper's uniform-step estimates, the two constructions tie in leading order.*

If we use the improved two-egg strategy highlighted in the talk

The talk explicitly presents the triangular-number/decreasing-step strategy, which changes the two-egg leading constant from 2 to $\sqrt{2}$. Then the 2-2 construction improves to $2\sqrt{2} \approx 2.82843$. But the 1-3 construction can use the exact three-egg asymptotic as well, giving 2.74636. Hence the fair comparison of the best subroutines is:

1-3 wins: $2.74636 < 2.82843$.

6. Why the answer is not the exact four-egg optimum

The result above compares the two *structured strategies* in the question. It is not a claim that either is globally optimal among all possible four-egg decision trees.

The paper records the stronger dynamic-counting fact that d drops with k eggs can handle

$$\sum_{j=1}^{k} \binom{d}{j}$$

floors. For $k = 4$ and large d, the leading term is $d^4/24$, so the unrestricted optimum has asymptotic scale

$$d \sim (24N)^{1/4} \approx 2.21336\, N^{1/4},$$

which is better than either structured strategy. This is exactly why the constant-factor question is interesting: forcing a strategy into a particular recursive architecture costs something.

7. A numerical check

For a concrete large value, take $N = 10^8$.

The asymptotic formulas predict

$$2.74636\,(10^8)^{1/4} \approx 274.6, \qquad 2.82843\,(10^8)^{1/4} \approx 282.8.$$

If the outer jump/block sizes are restricted to integers and the inner two- and three-egg problems are solved exactly, a direct integer check gives approximately

$$\text{1-3: 275 drops}, \qquad \text{2-2: 282 drops},$$

with near-optimal choices of $S \approx 1.45\times10^6$ and $F \approx 9.45\times10^3$, respectively. The finite-N calculation is already very close to the asymptotic prediction.

8. Takeaway

The cleanest conclusion is:

$$\boxed{\begin{array}{ll} \text{1-3:} & 4(2/9)^{1/4}N^{1/4} \approx 2.74636N^{1/4}, \\ \text{2-2:} & 2\sqrt{2}\,N^{1/4} \approx 2.82843N^{1/4}. \end{array}}$$

Thus the **1-3 strategy is asymptotically better by about 2.9%**, provided that the three-egg subproblem is solved using its optimal dynamic egg-drop asymptotic. Under the simpler fixed-step estimates developed in the paper, the two approaches tie at the leading constant 4; the improvement comes from using the triangular-number strategy for two eggs and, consistently, the exact dynamic capacity for three eggs.

Sources. The terminology and recursive fixed-step strategy follow Xiangwen Cao, Zongyun Chen, and Steven J. Miller, "Egg Drop Problems: They Are All They Are Cracked Up To Be!" (Fibonacci Quarterly, 2026; the uploaded copy is dated May 10, 2026). The paper's Theorem 2.1 gives the fixed-step bound $P_1(k) \le \lceil kN^{1/k} \rceil$ and its introduction records Boardman's exact dynamic capacity $\sum_{j=1}^{k} \binom{n}{j}$. The July 28, 2026 talk explicitly develops the decreasing-step/triangular strategy for two eggs and the recursive k-egg viewpoint.

Talk: https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/eggdropmath_Hampshire28Jul2026pdf

Paper: https://web.williams.edu/Mathematics/sjmiller/public_html/math/papers/EggDropProblems_FibonacciQuarterly.pdf

Video: https://youtu.be/98cpdt2EsCM